Determinant

89. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$x + 3y + 7z = 0$

$-x + 4y + 7z = 0$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ has a non-trivial solution, is:

(A) three (B) four (C) two (D) one

$\theta \in (0, \pi)$ के ऐसे मानों की संख्या, जिनके लिए निम्न रैखिक समीकरण निकाय

$x + 3y + 7z = 0$

$-x + 4y + 7z = 0$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ का एक अतुच्छ हल है, है:

(A) तीन (B) चार (C) दो (D) एक

Ans. C

Sol. for non trivial solution

$$\begin{vmatrix} 1 & 3 & 7 \\ -1 & 4 & 7 \\ \sin 3\theta & \cos 2\theta & 2 \end{vmatrix} = 0$$

$$\sin 3\theta(21-28) - \cos 2\theta(7+7) + 2(4+3) = 0$$

$$-7\sin 3\theta - 14\cos 2\theta + 14 = 0$$

$$-\left[3\sin\theta - 4\sin^3\theta\right] - 2\left(1 - 2\sin^2\theta\right) + 2 = 0$$

$$4\sin^3\theta + 4\sin^2\theta - 3\sin\theta = 0$$

$$\sin\theta\left[4\sin^2\theta + 6\sin\theta - 2\sin\theta - 3\right] = 0$$

$\sin\theta = 0$ $\qquad 2\sin\theta(2\sin\theta + 3) - (2\sin\theta + 3) = 0$

$$(2\sin\theta - 1)(2\sin\theta + 3) = 0$$

$$\sin\theta = \frac{1}{2} \qquad \sin\theta = \frac{-3}{2} \text{ Not possible}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

90. Let $a_1, a_2, a_3, \ldots\ldots\ldots a_{10}$ be in G.P. with $a_i > 0$ for i = 1, 2, ……,10 and S be the set of pairs (r, k), $k \in N$ (the set of natural numbers) for which $\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$

Then the number of elements in S, is:

(A) 4 (B) 10 (C) 2 (D) Infinity many

माना $a_1, a_2, a_3, \ldots\ldots\ldots a_{10}$ गुणोत्तर श्रेढी में है तथा i = 1, 2, ……,10 के लिए $a_i > 0$ है और S ऐसे युग्मों (r, k), $k \in N$

(प्राकृत संख्याओं का समुच्चय) का समुच्चय है जिनके लिए $\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$

तो S के अवयवों की संख्या है

(A) 4 (B) 10 (C) 2 (D) अनन्त

Ans. D

Sol. $\therefore\ a_1,\ a_2 \ldots\ldots\ldots a_{10} \to \text{GP}$

$\therefore\ \ln a_1,\ \ln a_2,\ \ln a_3 \ldots\ldots\ldots \ln a_{10} \to \text{AP}$

$$\begin{vmatrix} r\ln a_1 + k\ln a_2 & r\ln a_2 + k\ln a_3 & r\ln a_3 + k\ln a_4 \\ r\ln a_4 + k\ln a_5 & r\ln a_5 + k\ln a_6 & r\ln a_6 + k\ln a_7 \\ r\ln a_7 + k\ln a_8 & r\ln a_8 + k\ln a_9 & r\ln a_9 + k\ln a_8 \end{vmatrix} = 0$$

Apply $C_1 \to C_1 - C_2,\ C_2 \to C_2 - C_3$

Now, $C_1 = C_2$

So determinant = zero

91. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)\,(x+a+b+c)^2$, $x \neq 0$ and $a+b+c \neq 0$, then x is equal to

(A) abc (B) –(a+b+c) (C) –2(a+b+c) (D) 2(a+b+c)

यदि $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)\,(x+a+b+c)^2$, $x \neq 0$ तथा $a+b+c \neq 0$, तो x बराबर है–

(A) abc (B) –(a+b+c) (C) –2(a+b+c) (D) 2(a+b+c)

Ans. C

Sol. $R_1 \longrightarrow R_1 + R_2 + R_3$

$$(a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$c_1 \to c_1 \to c_2 \qquad c_2 \to c_2 \to c_3$

$$(a+b+c)\begin{vmatrix} 0 & 0 & 1 \\ a+b+c & -(a+b+c) & 2b \\ 0 & a+b+c & c-a-b \end{vmatrix}$$

$(a+b+c)\left[1(a+b+c)\right]^2=(a+b+c)(x+a+b+c)^2$

$x+a+b+c=\pm(a+b+c)$

$x=-2\,(a+b+c)$

92. An ordered pair (α,β) for which the system of linear equations :

$(1+\alpha)x+\beta y+z=2$

$\alpha x+(1+\beta)y+z=3$

$\alpha x+\beta y+2z=2$

has a unique solution, is :

(A) (–4, 2) (B) (2,4) (C) (–3,1) (D) (1,–3)

एक ऐसा क्रमित युग्म (α,β) जिसके लिए रैखिक समीकरण निकाय

$(1+\alpha)x+\beta y+z=2$

$\alpha x+(1+\beta)y+z=3$

$\alpha x+\beta y+2z=2$

का एकमात्र एक हल है, हैं –

(A) (–4, 2) (B) (2,4) (C) (–3,1) (D) (1,–3)

Ans. B

Sol. For unique solution $\Delta\neq 0$

$$\begin{vmatrix}1+\alpha & \beta & 1\\ \alpha & 1+\beta & 1\\ \alpha & \beta & 2\end{vmatrix}\neq 0$$

$$\begin{vmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ \alpha & \beta & 2\end{vmatrix}\neq 0$$

$1(2+\beta)+1(+\alpha)\neq 0$

$\alpha+\beta+2\neq 0$

93. The system of linear equations :

$x+y+z=2$

$2x+3y+2z=5$

$2x+3y+(a^2-1)z=a+1$

(A) Is inconsistent when $|a|=\sqrt{3}$ (B) Has a unique solution for $|a|=\sqrt{3}$

(C) Has infinitely many solution for a = 4 (D) is inconsistent when a = 4

रैखिक समीकरण निकाय

$x+y+z=2$

$2x+3y+2z=5$

$2x + 3y + (a^2 - 1)z = a + 1$

(A) असंगत है जब $|a| = \sqrt{3}$ (B) का $|a| = \sqrt{3}$ के लिए अद्वितीय हल है।

(C) के a = 4 के लिए अनन्त हल हैं। (D) असंगत है जब a = 4

Ans. A

Sol. $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 2 & 3 & (a^2-1) \end{vmatrix} = 0$

$\Rightarrow |a| = \sqrt{3}$

$\Delta_x \neq 0$ for $a = \sqrt{3}$ so system of equations are inconsistent

94. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z), then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to:

(A) $\frac{1}{2}$ (B) -4 (C) $\frac{3}{4}$ (D) $-\frac{1}{4}$

यदि समीकरण निकाय $2x + 3y - z = 0$, $x + ky - 2z = 0$ तथा $2x - y + z = 0$ का एक अतुच्छ (non-trivial) हल (x, y, z) है, तो $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ बराबर है :

(A) $\frac{1}{2}$ (B) -4 (C) $\frac{3}{4}$ (D) $-\frac{1}{4}$

Ans. A

Sol. $\Delta = 0 \Rightarrow \begin{vmatrix} 2 & 3 & -1 \\ 1 & k & -2 \\ 2 & -1 & 1 \end{vmatrix} = 0 \Rightarrow \boxed{k = \frac{9}{2}}$

$\therefore$ Equations are $2x + 3y - z = 0$ ....(i)

$2x - y + z = 0$ ....(ii)

$2x + 9y - 4z = 0$ ....(iii)

By (i) – (ii) $\boxed{2y = z}$ $\therefore$ $\boxed{z = -4x}$ and $\boxed{2x + y = 0}$

$\therefore \frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k = \frac{-1}{2} + \frac{1}{2} - 4 + \frac{9}{2}$

$= \frac{1}{2}$

95. If $\Delta_1 = \begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; then for all $\theta \in \left(0, \frac{\pi}{2}\right)$:

(A) $\Delta_1 + \Delta_2 = -2x^3$ (B) $\Delta_1 - \Delta_2 = -2x^3$

(C) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$ (D) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$

यदि $\Delta_1 = \begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ तथा $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; तो सभी $\theta \in \left(0, \frac{\pi}{2}\right)$ के लिए :

(A) $\Delta_1 + \Delta_2 = -2x^3$ (B) $\Delta_1 - \Delta_2 = -2x^3$

(C) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$ (D) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$

Ans. A

Sol. $\Delta_1 = \begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$

$= x(-x^2 - 1) - \sin\theta(-x\sin\theta - \cos\theta) + \cos\theta(-\sin\theta + x\cos\theta)$

$= -x^3 - x + x\sin^2\theta + \sin\theta\cos\theta - \cos\theta\sin\theta + x\cos^2\theta$

$= -x^3 - x + x = -x^3$

Similarly $\Delta_2 = -x^3$

$\Delta_1 + \Delta_2 = -2x^3$

96. The sum of the real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, is equal to :

(A) 6 (B) 0 (C) –4 (D) 1

समीकरण $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, के वास्तविक मूलों का योगफल है :

(A) 6 (B) 0 (C) –4 (D) 1

Ans. B

Sol. $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$

$\Rightarrow x(-3x^2 - 6x - 2x^2 + 6x) - 6(-3x + 9 - 2x - 4) - (4x - 9xA) = 0$
$\Rightarrow x(-5x^2) - 6(-5x + 5) - 4x + 9x = 0$
$\Rightarrow x^3 - 7x + 6 = 0$
All the roots are real

$\therefore$ Sum of real roots $= \frac{0}{1} = 0$

97. The system of linear equations

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10$ has :

(A) a unique solution when $\lambda = -8$ (B) no solution when $\lambda = 2$

(C) no solution when $\lambda = 8$ (D) infinitely many solutions when $\lambda = 2$

रेखिक समीकरण निकाय

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10$ है :

(A) का मात्र एक हल है जब $\lambda = -8$ (B) का कोई हल नहीं है जब $\lambda = 2$

(C) का कोई हल नहीं है जब $\lambda = 8$ (D) के अनन्त हल हैं जब $\lambda = 2$

Ans. B

Sol. $D = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix} = (\lambda + 8)(2 - \lambda)$

For unique solution $\lambda \neq 2, -8$
for $\lambda = 2$

$D_x = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix}$

$D_x = 16 \neq 0$ (No solution)

98. If for some α and β in R, the intersection of the following three planes

$x + 4y - 2z = 1$

$x + 7y - 5z = \beta$

$x + 5y + \alpha z = 5$

is a line in $\mathbf{R}^3$, then $\alpha + \beta$ is equal to :

(A) 2 (B) –10 (C) 0 (D) 10

यदि R में किन्ही α तथा β के लिए निम्न तीन समतलों

$x + 4y - 2z = 1$

$x + 7y - 5z = \beta$

$x + 5y + \alpha z = 5$

का प्रतिच्छेदन, $\mathbf{R}^3$ में एक रेखा है, तो $\alpha + \beta$ का मान है :

(A) 2 (B) –10 (C) 0 (D) 10

Ans. D

Sol. $x + 4y - 2z = 1$

$x + 7y - 5z = \beta$

$x + 5y + \alpha z = 5$

$\Rightarrow$ System has infinite solutions

$$D = \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$1(7\alpha + 25) - 4(\alpha + 5) - 2(5 - 7) = 0$

$\Rightarrow$ $7\alpha + 25 - 4\alpha - 20 + 4 = 0$

$3\alpha + 9 = 0$

$\therefore \alpha = -3$

$$D_x = \begin{vmatrix} 1 & 4 & -2 \\ \beta & 7 & -5 \\ 5 & 5 & -3 \end{vmatrix} = 0$$

$\Rightarrow 1(-21 + 25) - 4(-3\beta + 25) - 2(5\beta - 35) = 0$

$\Rightarrow 4 + 12\beta - 100 - 10\beta + 70 = 0$

$2\beta - 26 = 0$

$\beta = 13$

$$D_y = \begin{vmatrix} 1 & 1 & -2 \\ 1 & \beta & -5 \\ 1 & 5 & -3 \end{vmatrix} = 0$$

$\Rightarrow 1(-3\beta + 25) - 1(-3 + 10) + 1(-5 + 2\beta) = 0$

$-3\beta + 25 - 7 - 5 + 2\beta = 0$

$-\beta + 13 = 0$

$\therefore\ \beta = 13$

$$D_z = \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$\Rightarrow 1(5-7) - \beta(5-4) + 5(7-4) = 0$
$-2 - \beta + 15 = 0$
$\beta = 13$

$\alpha + \beta = -3 + 13 = 10$

99. Let $a - 2b + c = 1$.

If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then :

(A) $f(50) = 1$ (B) $f(50) = -501$ (C) $f(-50) = -1$ (D) $f(-50) = 501$

माना Let $a - 2b + c = 1$है। यदि

If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$ है, तो :

(A) $f(50) = 1$ (B) $f(50) = -501$ (C) $f(-50) = -1$ (D) $f(-50) = 501$

Ans. A

Sol. $a - 2b + c = 1$

$$f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$$

$R_1 \to R_1 + R_3 - 2R_2$

$$f(x) = \begin{vmatrix} 2x+a+c-2x-2b & 2x+6-2x-6 & 2x+4-2x-4 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$$

$$f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$$

$= 1((x+3)^2 - (x+4)(x+2)$
$= x^2 + 6x + 9 - x^2 - 6x - 8 = 1$

100. The following system of linear equations

$7x + 6y - 2z = 0$

$3x + 4y + 2z = 0$

$x - 2y - 6z = 0$, has

(A) only the trivial solution

(B) infinitely many solutions, (x, y, z) satisfying $x = 2z$

(C) infinitely many solutions, (x, y, z) satisfying $y = 2z$

(D) no solution

रैखिक समीकरणों के निम्न निकाय

$7x + 6y - 2z = 0$

$3x + 4y + 2z = 0$

$x - 2y - 6z = 0$

(A) का केवल तुच्छ हल हैं।

(B) $x = 2z$ को सन्तुष्ट करने वाले अनन्त हल (x, y, z) हैं।

(C) $y = 2z$ को सन्तुष्ट करने वाले अनन्त हल (x, y, z) हैं।

(D) का होई हल नहीं है।

Ans. B

Sol. $7x + 6y - 2z = 0$

$3x + 4y + 2z = 0$

$x - 2y - 6z = 0$

$$D = \begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix} = 7(-24+4) - 6(-18-2) - 2(-6-4)$$

$= 7(-20) - 6(-20) - 2(-10)$

$= -140 + 120 + 20 = 0$

$$\begin{array}{l} 7x + 6y = 2t \\ 3x + 4y = -2t \\ \hline 21x + 18y = 6t \\ 21x + 28y = -14t \\ \hline -10y = 20t \end{array} \quad \Rightarrow y = -2t$$

$7x + 6(-2t) = 2t$

$7x = 14t \Rightarrow x = 2t$

$x = 2z$

101. If $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} =$

$Ax^3 + Bx^2 + Cx + D$, then $B + C$ is equal to :

यदि $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} =$

$Ax^3 + Bx^2 + Cx + D$ है, तो $B + C$ is बराबर है :

(A) –3 (B) 1 (C) 9 (D) –1

Ans. A

Sol. $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & -x+2 & -2x+3 \\ 1 & -x+1 & x-5 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$

$C_2 \to C_2 - 2C_1$, $C_3 \to C_3 - 3C_1$

$\begin{vmatrix} x-2 & 1 & 2 \\ 1 & -x & -2x \\ 1 & -x-1 & x-8 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$

$-3x^3 + 12x^2 - 15x + 6 = Ax^3 + Bx^2 + Cx + D$

$B = 12$, $C = -15$

$B + C = -3$

102. Let S be the set of all integer solutions, (x, y, z) of the system of equations

$x - 2y + 5z = 0$

$-2x + 4y + z = 0$

$-7x + 14y + 9z = 0$

such that $15 \le x^2 + y^2 + z^2 \le 150$. Then, the number of elements in the set S is equal to _____.

यदि समीकरण निकाय

$x - 2y + 5z = 0$

$-2x + 4y + z = 0$

$-7x + 14y + 9z = 0$

के पूर्णांकीय हलों (x, y, z) का समुच्चय S है, जिनके लिए $15 \le x^2 + y^2 + z^2 \le 150$,तो S के अवयवों की संख्या है _____।

Ans. 8

Sol. $x - 2y + 5z = 0$

$-2x + 4y + z = 0$

$-7x + 14y + 9z = 0$

$$D = \begin{vmatrix} 1 & -2 & 5 \\ -2 & 4 & 1 \\ -7 & 14 & 9 \end{vmatrix}$$

$D = 1(36 - 14) + 2(-18 + 7) + 5(-28 + 28)$

$D = 22 - 22 + 0 = 0$

Let $y = k$

$x - 2k + 5z = 0$

$-2x + 4k + z = 0$

$z = 0$

$x = 2k$

solutions

(x, y , z)

(2k, k, 0)

$s \le x^2 + y^2 + z^2 \le 150$

$s \le 4k^2 + k^2 + 0^2 \le 150$

$1 \le k^2 \le 30$

$K = \pm 5, \pm 4, \pm 3, \pm 2$

hence total number of solutions 8.

103. If the minimum and the maximum values of the function $f : \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \to R$, defined by

$$f(\theta) = \begin{vmatrix} -\sin^2\theta & -1-\sin^2\theta & 1 \\ -\cos^2\theta & -1-\cos^2\theta & 1 \\ 12 & 10 & -2 \end{vmatrix}$$ are m and M respectively, then the ordered pair (m, M) is equal to :

यदि $f(\theta)=\begin{vmatrix}-\sin^2\theta & -1-\sin^2\theta & 1\\ -\cos^2\theta & -1-\cos^2\theta & 1\\ 12 & 10 & -2\end{vmatrix}$ द्वारा परिभाषित फलन $f:\left[\frac{\pi}{4},\frac{\pi}{2}\right]\to R$ के निम्नतम तथा उच्चतम मान क्रमशः m तथा M हैं, तो क्रमित युग्म (m, M) बराबर हैं :

(A) $(0,2\sqrt{2})$ (B) (0, 4) (C) (–4, 4) (D) (–4, 0)

Ans. D

Sol. $C_2\to C_2-C_1+C_3$

$$f(\theta)=\begin{vmatrix}-\sin^2\theta & 0 & 1\\ -\cos^2\theta & 0 & 1\\ 12 & -4 & -2\end{vmatrix}=4\cos 2\theta$$

$\theta\in\left[\frac{\pi}{4},\frac{\pi}{2}\right]$

$\Rightarrow 4\cos 2\theta\in[-4,0]$

104. If $a+x=b+y=c+z+1$, where a, b, c, x, y, z are non-zero dinstinct real numbers, then $\begin{vmatrix}x & a+y & x+a\\ y & b+y & y+b\\ z & c+y & z+c\end{vmatrix}$ is equal to :

यदि $a+x=b+y=c+z+1$ हैं, जहाँ a, b, c, x, y, z शून्येत्तर भिन्न वास्तविक संख्याएँ हैं, तो $\begin{vmatrix}x & a+y & x+a\\ y & b+y & y+b\\ z & c+y & z+c\end{vmatrix}$ बराबर है :

(A) 0 (B) y (b – a) (C) y (a – b) (D) y (a – c)

Ans. C

Sol. $C_3\to C_3-C_1$

$$=\begin{vmatrix}x & a+y & a\\ y & b+y & b\\ z & c+y & c\end{vmatrix}$$

$C_2 \to C_2 - C_3$

$$= \begin{vmatrix} x & x & a \\ y & y & b \\ z & y & c \end{vmatrix}$$

$$= y\begin{vmatrix} x & 1 & a \\ y & 1 & b \\ z & 1 & c \end{vmatrix}$$

$R_1 \to R_1 - R_2$, then $R_2 \to R_2 - R_3$

$$= y\begin{vmatrix} x-y & 0 & a-b \\ y-z & 0 & b-c \\ z & 1 & c \end{vmatrix}$$

$$= y(-1)[(x-y)(b-c)-(y-z)(a-b)]$$

$$= -y\big((b-c)(b-a)-(a-b)(c-b-1)\big)$$

$$= y(a-b)$$

105. If the system of linear equations

$x + y + 3z = 0$

$x + 3y + k^2z = 0$

$3x + y + 3z = 0$

has a non-zero solution (x, y, z) for some $k \in R$, then $x + \left(\frac{y}{z}\right)$ is equal to :

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$x + y + 3z = 0$

$x + 3y + k^2z = 0$

$3x + y + 3z = 0$

का, किसी $k \in R$, के लिए , एक शून्येत्तर हल (x, y, z) है, तो $x + \left(\frac{y}{z}\right)$ बराबर है :

(A) – 9 (B) – 3 (C) 3 (D) 9

Ans. B

Sol. For non-zero solutiuon

$D = 0$

$$\begin{vmatrix} 1 & 1 & 3 \\ 1 & 3 & k^2 \\ 3 & 1 & 3 \end{vmatrix} = 0$$

$1(9 - k^2) - (3 - 3k^2) + 3(1 - 9) = 0$

$9 - k^2 - 3 + 3k^2 - 24 = 0$

$2k^2 - 18 = 0$

$k^2 = 9$

system of equation

$x + y + 3z = 0$ .....(1)

$x + 3y + 9z = 0$ .....(2)

$3x + y + 3z = 0$ .....(3)

(A) – (3)

$2x = 0, \Rightarrow x = 0$

$\frac{y}{z} = -3$

106. Let m and M be respectively the minimum and maximum values of

$$\begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x \end{vmatrix}$$

Then the ordered pair (m, M) is equal to :

माना m तथा M

$$\begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1+\cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1+\sin 2x \end{vmatrix}$$ के, क्रमशः

न्यूनतम तथा अधिकतम मान है, तो क्रमित युग्म (m, M) बराबर है :

(A) (1, 3) (B) (– 3, 3) (C) (– 4, – 1) (D) (– 3, – 1)

Ans. D

Sol. $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$D = \begin{vmatrix} \cos^2 x & 1+\sin^2 x & \sin 2x \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$D = -2 - \sin 2x$

$m = -3$

$M = -1$

107. The sum of distinct values of λ for which the system of equations

$(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$

$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$

$2x+(3\lambda+1)y+3(\lambda-1)z=0$

has non-zero solutions, is _____.

λ के उन भिन्न मानों का योग, जिनके लिए समीकरण निकाय

$(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$

$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$

$2x+(3\lambda+1)y+3(\lambda-1)z=0$

के शून्येतर (non-zero) हल हैं, है_____।

Ans. 3

Sol. $(\lambda-1)x+(3\lambda+1)y+2\lambda z=0$

$(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0$

$2x+(3\lambda+1)y+(3\lambda-3)z=0$

$$\begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$R_1 \to R_1 - R_2$

$$\begin{vmatrix} 0 & -\lambda+3 & +\lambda-3 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$(\lambda-3)\begin{vmatrix} 0 & -1 & 1 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$\lambda = 3 \text{ or } \begin{vmatrix} 0 & -1 & 1 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$C_2 \to C_2 + C_3$

$$\begin{vmatrix} 0 & 0 & 1 \\ \lambda-1 & 5\lambda+1 & \lambda+3 \\ 2 & 6\lambda-2 & 3\lambda-3 \end{vmatrix} = 0$$

$$\big((\lambda-1)(6\lambda-2)-2(5\lambda+1)\big) = 0$$

$$6\lambda^2 - 8\lambda + 2 - 10\lambda - 2 = 0$$

$$\lambda^2 - 3\lambda = 0$$

$$\lambda(\lambda-3) = 0$$

$\Rightarrow \lambda = 0, 3$ sum = 0 + 3 = 3

108. For the system of linear equations :

$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R$

Consider the following statements :

(A) The system has unique solution if $k \neq 2, k \neq -2$.

(B) The system has unique solution if $k = -2$.

(C) The system has unique solution if $k = 2$.

(D) The system has no–solution if $k = 2$.

(E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct ?

(A) (C) and (D) only (B) (A) and (E) only

(C) (B) and (E) only (D) (A) and (D) only

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$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in R$

के लिए, नीचे दिए कथनों पर विचार कीजिए :

(A) निकाय का केवल एक हल है, यदि $k \neq 2, k \neq -2$ है।

(B) निकाय का केवल एक हल है, यदि $k = -2$ है।

(C) निकाय का केवल एक हल है, यदि $k = 2$ है।

(D) निकाय का कोई हल नहीं है, यदि $k = 2$ है।

(E) निकाय के अनन्त हल हैं, यदि $k \neq -2$ है।

तो निम्न कथनों में कौन से सत्य हैं ?

(A) केवल (C) तथा (D)

(B) केवल (A) तथा (E)

(C) केवल (B) तथा (E)

(D) केवल (A) तथा (D)

Ans. D

Sol. $D = \begin{vmatrix} 1 & -2 & 0 \\ 1 & -1 & k \\ 0 & k & 4 \end{vmatrix} = 4 - k^2 = (2 - k)(2 + k)$

$$D_x = \begin{vmatrix} 1 & -2 & 0 \\ -2 & -1 & k \\ 6 & k & 4 \end{vmatrix} = -(k + 10)(k + 2)$$

$D \neq 0 \Rightarrow k \neq \pm 2$

$\Rightarrow$ system will have unique solution

$\Rightarrow$ A is true; E, B, C are false

If $k = 2 \Rightarrow D = 0$ $D_x \neq 0$

$\Rightarrow$ System will have no solution $\Rightarrow$ D is true

$\Rightarrow$ A & D are true.

109. The following system of linear equations :

$2x + 3y + 2z = 9$

$3x + 2y + 2z = 9$

$x - y + 4z = 8$

(A) Has infinitely many solutions

(B) Has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^2 = 12$

(C) Does not have any solution

(D) Has a unique solution

रैखिक समीकरण निकाय

$2x + 3y + 2z = 9$

$3x + 2y + 2z = 9$

$x - y + 4z = 8$

(A) के अनन्त हल हैं

(B) का एक हल (α, β, γ) है जो $\alpha + \beta^2 + \gamma^2 = 12$ को संतुष्ट करता है

(C) का कोई हल नहीं है

(D) का केवल एक हल है

Ans. D

Sol. $\Delta = \begin{vmatrix} 2 & 3 & 2 \\ 3 & 2 & 2 \\ 1 & -1 & 4 \end{vmatrix}$

$= 2(8 + 2) - 3(12 - 2) + 2(-3 - 2)$

$= 32 - 30 - 10 \qquad = -8$

$D \neq 0$

has a unique solution

110. Consider the following system of equations :

$x + 2y - 3z = a$

$2x + 6y - 11z = b$

$x - 2y + 7z = c,$

where a, b and c are real constants. Then the system of equations :

(A) has no solution for all a, b and c

(B) has infinite number of solutions when $5a = 2b + c$

(C) has a unique solution when $5a = 2b + c$

(D) has a unique solution for all a, b and c

निम्न समीकरण निकाय पर विचार कीजिए

$x + 2y - 3z = a$

$2x + 6y - 11z = b$

$x - 2y + 7z = c,$

जहाँ a, b तथा c वास्तविक अचर हैं। तो इस समीकरण निकाय :

(A) का सभी a, b तथा c के लिए कोई हल नहीं हैं (B) के अनन्त हल हैं जब $5a = 2b + c$

(C) का केवल एक हल है जब $5a = 2b + c$ (D) का सभी a, b तथा c के लिए केवल एक हल है

Ans. B

Sol. $D = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{vmatrix}$

$= 1(42 - 22) - 2(14 + 11) - 3(-4 - 6)$

$= 20 - 50 + 30 = 0$

$D_x = \begin{vmatrix} a & 2 & -3 \\ b & 6 & -11 \\ c & -2 & 7 \end{vmatrix}$

$= a[42 - 22] - b[14 - 6] + c(-22 + 18)$

$= 20a - 8b - 4c$

$= 4[5a - 2b - c]$

$D_y = \begin{vmatrix} 1 & a & -3 \\ 2 & b & -11 \\ 1 & c & 7 \end{vmatrix}$

$= -5[5a - 2b - c]$

$D_z = \begin{vmatrix} 1 & 2 & a \\ 2 & 6 & b \\ 1 & -2 & c \end{vmatrix}$

$= -2[5a - 2b - c]$

for $5a - 2b - c = 0$, $D_x = D_y = D_z = 0$

$\therefore$ system has infinite number of solutions

when $5a = 2b + c$

111. Let α, β, γ be the real roots of the equation $x^3 + ax^2 + bx + c = 0$, $(a, b, c \in R$ and $a, b \neq 0)$. If the system of equations (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$; $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is :

माना α, β, γ समीकरण $x^3 + ax^2 + bx + c = 0$, $(a, b, c \in R$ तथा $a, b \neq 0)$ के वास्तविक मूल हैं। यदि u, v, w में समीकरण निकाय $\alpha u + \beta v + \gamma w = 0$; $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ का अतुच्छ हल है, तो $\frac{a^2}{b}$ का मान है :

(A) 5 (B) 3 (C) 0 (D) 1

Ans. B

Sol. For the system of equations to have a non-trivial solution.

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$
$\Rightarrow \alpha = \beta = \gamma$ or $\alpha + \beta + \gamma = 0$
But since $\alpha + \beta + \gamma = -a$ and $a \neq 0$
$\Rightarrow \alpha + \beta + \gamma \neq 0$
So, $\alpha = \beta = \gamma$
So $3\alpha = -a$; $3\alpha^2 = b$

$$\frac{a^2}{b} = \frac{9\alpha^2}{3\alpha^2} = 3$$

112. Let S_1 and S_2 be respectively the sets of all $a \in R - \{0\}$ for which the system of linear equations :
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$
$(3a + 5)x + (a + 5)y + (a + 2)z = 3$
has unique solution and infinitely many solutions. Then :

(A) S_1 is an infinite set and $n(S_2) = 2$ (B) $S_1 = R - \{0\}$ and $S_2 = \phi$

(C) $n(S_1) = 2$ and S_2 is an infinite set (D) $S_1 = \phi$ and $S_2 = R - \{0\}$

माना सभी $a \in R - \{0\}$, जिनके लिए रैखिक समीकरण निकाय
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$

$(3a + 5)x + (a + 5)y + (a + 2)z = 3$

का केवल एक हल है तथा अनंत हल है, के समुच्चय क्रमशः S_1 तथा S_2 है। तो

(A) S_1 एक अपरिमित समुच्चय है तथा $n(S_2) = 2$ है

(B) $S_1 = R - \{0\}$ तथा $S_2 = \phi$

(C) $n(S_1) = 2$ तथा S_2 एक अपरिमित समुच्चय है

(D) $S_1 = \phi$ तथा $S_2 = R - \{0\}$

Ans. B

Sol. $$D = \begin{bmatrix} a & 2a & -3a \\ 2a+1 & 2a+3 & 2a+1 \\ 3a+5 & a+5 & a+2 \end{bmatrix} = a\left(15a^2 + 31a + 37\right)$$

$\Rightarrow D \neq a$ for all $a \in R - \{0\}$

113. If 1, $\log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x, then the value of the

determinant $\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ is equal to :

यदि एक वास्तविक संख्या x के लिए 1, $\log_{10}(4^x - 2)$ तथा $\log_{10}\left(4^x + \frac{18}{5}\right)$ एक समान्तर श्रेढ़ी में है, तो सारणिक

$\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ का मान बराबर _____।

Ans. 2

Sol. $$2\log_{10}\left(4^x - 2\right) = 1 + \log_{10}\left(4^x + \frac{18}{5}\right)$$

$$\log_{10}\left(4^x - 2\right)^2 = \log_{10}\left(10\left(4^x + \frac{18}{5}\right)\right)$$

$(4x - 2)^2 = 10.4^x + 36$ $\qquad$ $4^x = y$

$(y-2)^2 = 10y + 36$

$y^2 - 4y + 4 = 10y + 36$

$y^2 - 14y - 32 = 0$

$(y-16)(y+2) = 0$

$y = 16$

$y = -2$, which is not possible

$4^x = 16$

$x = 2$

Now, value of determinant

$$\begin{vmatrix} 3 & 1 & 4 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix}$$

$= 3(-2) - 1(0-4) + 4(1)$

$= -6 + 4 + 4$

$= 2$